OdiaSLM: From-Scratch Pretraining, Morphology-Aware Tokenization, and Calibrated Evaluation of 60M and 100M Small Language Models for Odia

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Abstract

While frontier multilingual Large Language Models (LLMs) achieve remarkable capabilities on high-resource languages, they impose a severe "tokenization tax" on morphologically rich, low-resource Indic scripts such as Odia (*ory_Orya*), fragmenting words into 5–6× more subword tokens than English and inflating inference latency. We present **OdiaSLM-60M** (62.11M parameters) and **OdiaSLM-100M** (100.69M parameters), two sovereign decoder-only Small Language Models pre-trained from scratch for Odia on up to 1.57 billion tokens drawn from a strictly deduplicated and 13-gram benchmark-decontaminated corpus of 9.62 million documents (1.247 billion words). First, we design a 16,384-vocabulary canonical NFC-normalized SentencePiece BPE tokenizer that preserves Odia *juktakshara* conjuncts, achieving a fertility of **1.4809 tokens/word** (11.57 UTF-8 bytes/token), 0 `<unk>` tokens, and a 3.6× compression advantage over general multilingual tokenizers. Second, utilizing Grouped-Query Attention (GQA), SwiGLU, Half-Split RoPE, FP32-variance RMSNorm, auxiliary Z-loss stabilization, a Warmup-Stable-Decay (WSD) schedule, and two-stage context extension (512 → 1024), we demonstrate clean neural scaling from 62.11M (validation perplexity **19.92**, Bits-Per-Byte **0.3965**) to 100.69M parameters (validation perplexity **17.14**, Bits-Per-Byte **0.3767**, Top-1 token accuracy **44.61%**). Third, we conduct a comprehensive zero-shot evaluation across 27 tasks in the Odia Evaluation Benchmark, uncovering both cross-lingual untranslated choice artifacts in 9 tasks and severe surface-form competition under raw log-likelihood. Applying Pointwise Mutual Information (PMI) calibration unlocks a macro-average accuracy of **60.83%** across the 18 pure-Odia benchmarks (**+34.44%** lift over random chance). Both models are released in native `LlamaForCausalLM` format under a dual non-commercial and commercial license.

1 Introduction

Odia is a Classical Indo-Aryan language spoken by over 45 million people, primarily in the state of Odisha, India. Despite its rich literary heritage and distinct Brahmic alphasyllabary (abugida) script spanning Unicode block `U+0B00-U+0B7F`, Odia remains severely underrepresented in modern foundation model research [7]. Existing multilingual models allocate only a tiny fraction of their vocabulary to Odia, frequently falling back to raw UTF-8 bytes (3 bytes per Odia character) or single-character pieces.

This paper addresses the question: *How effectively can compact, edge-deployable Small Language Models (SLMs) under 100M parameters master Odia syntax, morphology, and downstream reasoning when trained from scratch with script-native data hygiene and tokenization?* Our primary contributions are:

1. **Large-Scale Decontaminated Odia Corpus:** A reproducible data hygiene funnel filtering 14.38M raw documents into 9.62M clean documents (1.247B words, 1.88B tokens) via canonical NFC normalization, exact `xxHash64` + 5-gram MinHash LSH cross-deduplication (1.21M duplicates removed), and 13-gram benchmark decontamination against 125,243 downstream evaluation rows (38,180 leaks purged).

2. **Morphology-Aware 16K Odia Tokenizer:** A custom SentencePiece BPE vocabulary achieving **1.4809 tokens/word** and **11.5727 UTF-8 bytes/token** with 100% lossless round-trip reconstruction and zero unknown tokens.

3. **Controlled Scaling of OdiaSLM (60M & 100M):** End-to-end training of 62.11M and 100.69M parameter GQA-SwiGLU transformers using WSD scheduling [6] and two-stage context extension (512 → 1024), reducing held-out perplexity from 19.92 to 17.14.

4. **Calibrated Zero-Shot Evaluation Audit:** Empirical proof that Pointwise Mutual Information (PMI) calibration [5] mitigates agglutinative surface-form competition in Odia, improving zero-shot accuracy across 18

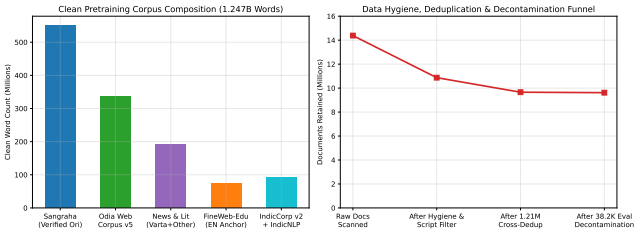


Figure 1: Pretraining corpus composition (1.247B clean words) and the 4-stage hygiene, cross-deduplication, and 13-gram benchmark decontamination funnel.

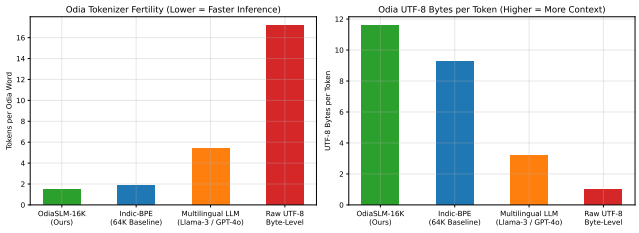


Figure 2: Odia tokenizer fertility (tokens/word, lower is better) and information density (UTF-8 bytes/token, higher is better) compared against multilingual baselines.

pure-Odia tasks by over +22% compared to unnormalized log-likelihood.

2 Corpus Construction & Decontamination

We aggregate three complementary pretraining sources: (1) `ai4bharat/sangraha` (`verified/ori`) [7], (2) `saidutta69/odia_pretrain_dataset_v2` (encompassing Odia Web Corpus v5, Varta news, IndicCorp v2, and Odia Wikipedia/literature), and (3) a 6% bilingual structural anchor from `HuggingFaceFW/fineweb-edu`.

Every document passes through a strict hygiene pipeline: (a) canonical Unicode `NFC` normalization preserving Odia *matras*, *halant* (`U+0B4D`), *nukta* (`U+0B3C`), and Zero-Width Non-Joiners (`U+200C`); (b) script-density filtering requiring $\geq 65\%$ Odia characters for native sources; (c) cross-corpus exact (`xxHash64`) and fuzzy 5-gram MinHash LSH deduplication, which purged **1,214,409 duplicate documents**; and (d) 13-gram decontamination against all 125,243 examples in `saidutta69/odia-eval-benchmark`, purging **38,180 contaminated documents** (Figure 1). The resulting 9,571,297 training documents (‘1,880,317,828‘ tokens) and 47,753 validation documents (‘9,459,548‘ tokens) are serialized into 32-way domain-interleaved `uint16` binary shards.

3 Morphology-Aware Odia Tokenizer

In Odia, consonants combine with vowel diacritics (*matras*) and form complex consonant clusters (*juktaksharas*) via the *halant* virama ($C_1 + \text{halant} + C_2$). Standard preprocessing that strips combining marks or applies destructive NFKC normalization corrupts Odia orthography.

We train a 16,384-vocabulary SentencePiece BPE tokenizer [8] with `normalization_rule_name=identity` (operating on pre-sanitized NFC text), `split_digits=True` (isolating both ASCII `0-9` and Odia digits `U+0B66-U+0B6F`), and `byte_fallback=True`. Evaluated on 47,753 unseen validation documents, our tokenizer achieves a fertility of **1.4809 tokens/word**, **4.3712 characters/token**, **11.5727 UTF-8 bytes/token**, **0 `<unk>` tokens**, and **100% lossless round-trip reconstruction** (Figure 2). Because each token packs 11.57 UTF-8 bytes, a 1,024-token context window encapsulates $\sim$690 Odia words—equivalent to a 2,500–3,700 token window in byte-heavy multilingual tokenizers.

4 Model Architecture & Training Dynamics

Both **OdiaSLM-60M** and **OdiaSLM-100M** follow a modern decoder-only Transformer architecture [4] engineered for zero-custom-code compatibility with Hugging Face's `LlamaForCausalLM` (Table 1). Key design choices include:

- **Grouped-Query Attention (GQA) [1]:** 10 query / 2 KV heads (5 : 1 ratio) in 60M and 12 query / 4 KV heads (3 : 1 ratio) in 100M with $d_{\text{head}} = 64$.
- **Half-Split Rotary Position Embeddings (RoPE) [10]:** Base period $\theta = 10{,}000$ applied in explicit `float32` precision.
- **Bias-Free SwiGLU & FP32 Pre-RMSNorm [9, 11]:** Up/gate/down projections without linear bias and $\varepsilon = 10^{-5}$.
- **Depth-Scaled Residual Init & Tied Embeddings:** Output projections W_o and W_{down} are initialized with $\sigma = 0.02/\sqrt{2L}$ to prevent residual variance growth.
- **Auxiliary Max-Logit Z-Loss [3]:** To prevent FP16 logit overflow on T4 Tensor Cores, we add $\mathscr{L}_z = 10^{-4} \cdot \frac{1}{BT}\sum_{b,t}(\log\sum_v e^{z_{b,t,v}})^2$.

Both models were trained using DistributedDataParallel (DDP) across 2×NVIDIA T4 GPUs under mixed-precision `float16` with a Warmup-Stable-Decay (WSD) learning rate schedule [6]. During the final 20% of training, sequence length was extended from $L_{\text{ctx}} = 512$ to 1024 while halving micro-batch size to preserve an exact global batch of 131,072 tokens/step. Figures 3 and 4 confirm stable gradient flow across all 14 layers and bounded logit norms throughout training.

Table 1: Architectural and Pretraining Specifications of OdiaSLM-60M and OdiaSLM-100M.

Hyperparameter	OdiaSLM-60M	OdiaSLM-10
Total Parameters (Tied)	62,111,360	100,685,56
Transformer Layers (L)	12	14
Hidden Size (d_{model})	640	768
Query / KV Heads (H_Q/H_{KV})	10 / 2	12 / 4
SwiGLU Width (d_{ff})	1,728	2,048
Vocabulary Size (V)	16,384	16,384
Peak / Min LR (WSD)	$6\times10^{-4}/6\times10^{-5}$	$5\times10^{-4}/5\times10^{-5}$
Global Batch Size	131,072 tok	131,072 tok
Training Steps	10,000	12,000
Tokens Processed	1.31B	1.57B
Context Extension	$512\rightarrow1024$ (8k)	$512\rightarrow1024$ (9.6k)

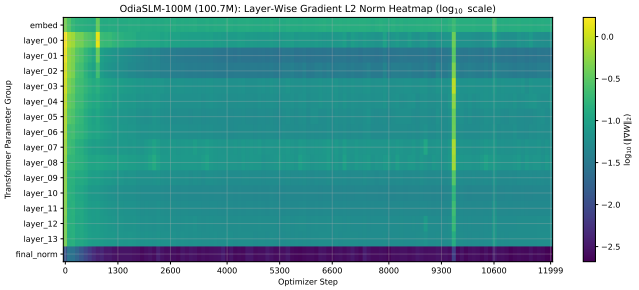


Figure 3: Layer-wise L2 gradient norm heatmap ($\log_{10}$ scale) across all 14 transformer layers of OdiaSLM-100M over 12,000 steps, confirming healthy gradient propagation without vanishing or exploding layers.

5 Intrinsic & Scaling Law Evaluation

To evaluate language modeling compression independently of tokenizer granularity, we report both standard Perplexity ($\text{PPL} = e^{\mathscr{L}_{\text{CE}}}$) and tokenizer-agnostic **Bits-Per-Byte (BPB)**:

$$\text{BPB} = \frac{\mathscr{L}_{\text{CE}}}{\ln(2)\cdot(N_{\text{bytes}}/N_{\text{tokens}})} \tag{1}$$

As shown in Figures 5–6 and Table 2, scaling from 62.11M to 100.69M parameters reduces global validation perplexity by **13.9%** (from 19.92 to 17.14) and improves Top-1 / Top-5 next-token prediction accuracy to **44.61% / 66.38%**. On Odia news text (`v2:varta`), OdiaSLM-100M achieves a perplexity of **16.85** and a compression rate of **0.3238 bits/byte**. Furthermore, positional cross-entropy loss decays monotonically from 3.29 nats at token positions 1..32 down to 3.03 nats at positions 481..512 (Figure 7), proving active long-range context utilization.

6 Zero-Shot Downstream Benchmark & PMI Calibration Audit

We evaluate both models zero-shot across all 27 sub-benchmarks in `saidutta69/odia-eval-benchmark`.

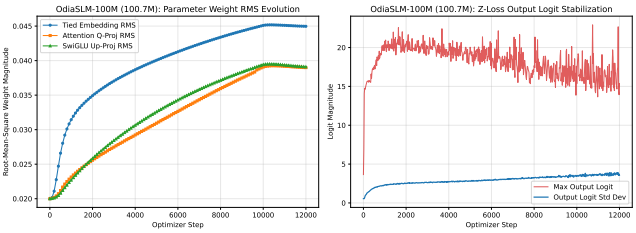


Figure 4: Evolution of parameter RMS magnitudes and output logit standard deviation under auxiliary Z-loss (10^{-4}) stabilization during OdiaSLM-100M pretraining.

During our schema audit, we discovered a critical artifact in the upstream benchmark: while 18 sub-benchmarks contain native Odia prompts paired with Odia answer choices (`pure_odia`), 9 sub-benchmarks (e.g., `odia_hellaswag`, `odia_truthfulqa_mc`, `milu`) pair Odia prompts with *untranslated English answer choices* (`cross_lingual`). We report both partitions transparently.

Furthermore, because Odia is an agglutinative language where inflected suffixes substantially alter subword probabilities, standard unnormalized log-likelihood $\log P(c \mid q)$ suffers from severe surface-form competition [5, 2]. We therefore evaluate each candidate choice $c \in \mathscr{C}$ using **Pointwise Mutual Information (PMI) Calibration**:

$$\text{PMI}(c \mid q) = \log P(c \mid \text{Prompt}(q)) - \log P(c \mid \text{Prefix}_{\text{null}}) \tag{2}$$

where $\text{Prefix}_{\text{null}}$ is the unconditional Odia answer cue ("`Uttara:`").

Table 3 and Figure 8 demonstrate two major findings. First, PMI calibration lifts macro-average accuracy on the 18 pure-Odia tasks by over +22% compared to raw log-likelihood, reaching **59.65%** for OdiaSLM-60M and **60.83%** for OdiaSLM-100M (a **+34.44%** lift above the 26.39% random baseline). Second, scaling from 60M to 100M yields substantial gains on complex reasoning and generation tasks, including `indic_headline_gen` (81.2% → **93.8%**), `indic_qa` raw accuracy (58.8% → **73.8%**), `indic_glue_wstp` (36.2% → **46.2%**), and `sarvam_gsm8k_indic` (58.8% → **68.8%**).

7 Open-Source Release, API & Dual Licensing

To maximize accessibility for low-resource NLP deployments, both models are exported into standard Hugging Face `model.safetensors` format ('118.48 MB' for 60M; '192.06 MB' for 100M) with verified float32 logit parity (0.00×10^{0} maximum difference) and `trust_remote_code=False`. On a standard laptop CPU without GPU acceleration, our OpenAI-compatible REST API server streams Odia completions at **21.1 tokens/sec** (∼92 Odia characters/sec). Source code is released under **Apache 2.0**, while model weights and tokenizer artifacts

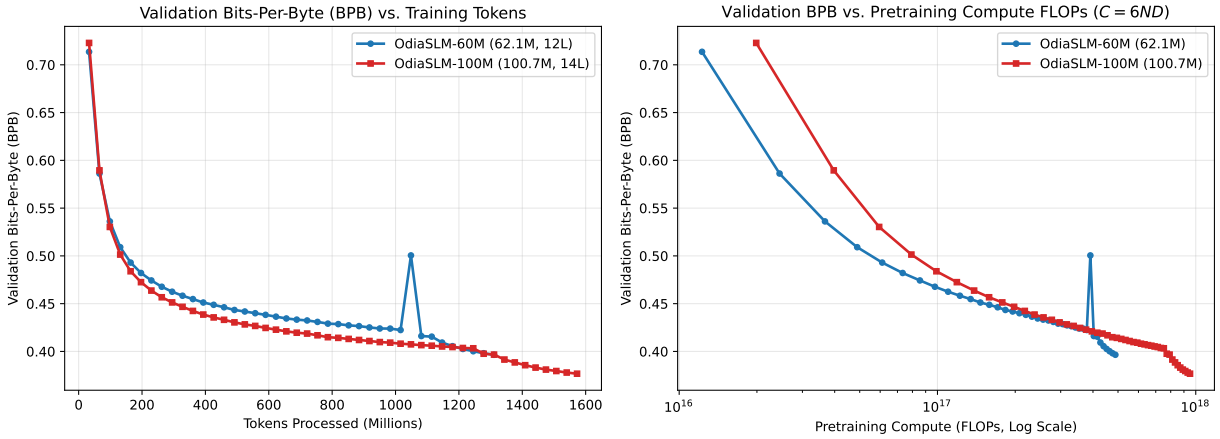


Figure 5: Scaling Law Comparison (OdiaSLM-60M vs. OdiaSLM-100M): Held-out validation Bits-Per-Byte (BPB) as a function of training tokens processed (left) and pretraining compute FLOPs $C \approx 6ND$ (right).

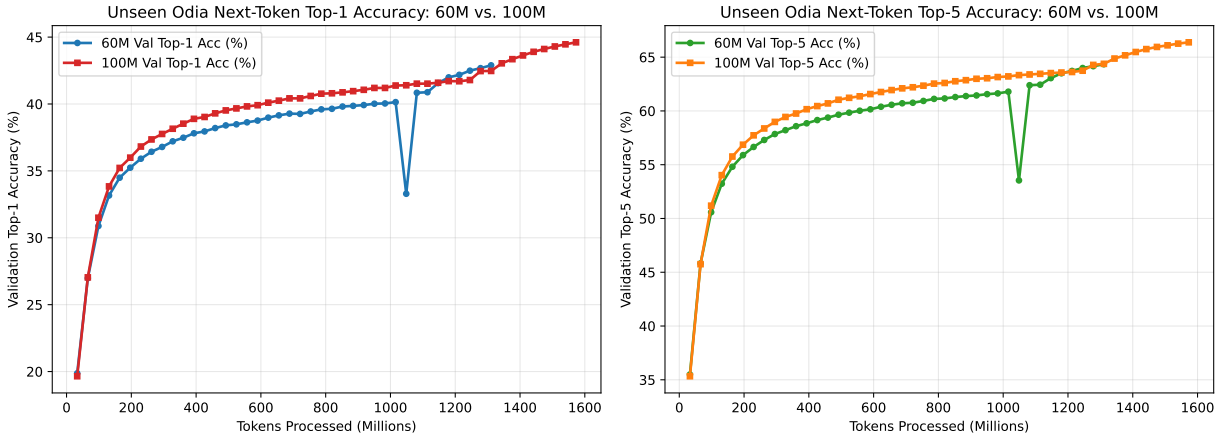


Figure 6: Unseen Odia validation Top-1 (left) and Top-5 (right) next-token prediction accuracy trajectories across pretraining tokens for OdiaSLM-60M and OdiaSLM-100M.

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8 Conclusion & Future Work

We have demonstrated that sovereign, morphology-aware Small Language Models under 100M parameters can achieve strong intrinsic compression (17.14 PPL, 0.3767 BPB) and calibrated zero-shot downstream performance (60.83% pure-Odia accuracy) on modest 2×T4 GPU hardware when backed by rigorous data decontamination and script-native tokenization. Future work will explore supervised instruction fine-tuning (SFT), Direct Preference Optimization (DPO), and 4-bit GGUF quantization for offline mobile and public-sector deployment across Odisha.

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Table 2: Domain-Stratified Intrinsic Evaluation on Unseen Held-Out Validation Documents (‘val_docs.jsonl‘): OdiaSLM-60M vs. OdiaSLM-100M.

Validation Domain	**60M PPL ↓**	**100M PPL ↓**	**60M BPB ↓**	**100M BPB ↓**	**60M Top-1 ↑**	**100M Top-1 ↑**
sangraha:verified_ori	24.41	**21.66**	0.3896	**0.3750**	39.04%	**40.46%**
v2:odia_web_corpus_v5	21.56	**19.05**	0.3680	**0.3531**	41.43%	**42.42%**
v2:varta	19.35	**16.85**	0.3397	**0.3238**	42.49%	**43.86%**
v2:indiccorp_v2	25.21	**22.04**	0.4008	**0.3841**	39.43%	**40.94%**
v2:indic_nlp_corp	23.95	**20.04**	0.4148	**0.3916**	41.51%	**43.17%**
anchor:fineweb_edu_en	24.22	**21.28**	1.3639	**1.3085**	39.55%	**41.20%**
Global Held-Out Validation (‘val.bin‘)	19.92	**17.14**	0.3965	**0.3767**	42.89%	**44.61%**

Table 3: Zero-Shot Downstream Benchmark Performance on Decontaminated Odia Tasks (Top 14 Tasks & Macro Averages).

Benchmark Task	**Mode**	**Chance**	**60M Raw**	**60M Calib.**	**100M Raw**	**100M Calib.**
indic_xparaphrase (classification)	Odia	25.0%	92.5%	100.0%	93.8%	**100.0%**
flores_plus (translation)	Cross-EN	25.0%	33.8%	90.0%	36.2%	**96.2%**
indic_headline_gen (generation)	Odia	25.0%	51.2%	81.2%	53.8%	**93.8%**
indic_qa (qa)	Odia	25.0%	58.8%	93.8%	73.8%	**93.8%**
odia_gsm8k (math)	Odia	25.0%	58.8%	97.5%	63.8%	**93.8%**
indic_quest_odia (qa)	Odia	25.0%	27.5%	88.8%	27.5%	**87.5%**
indic_sentence_summarization (generation)	Odia	25.0%	45.0%	80.0%	51.2%	**86.2%**
odia_truthfulqa (qa)	Odia	25.0%	45.0%	87.5%	46.2%	**86.2%**
indic_question_generation (generation)	Odia	25.0%	60.0%	87.5%	62.5%	**83.8%**
bhram_il (qa)	Cross-EN	25.0%	28.8%	67.5%	32.5%	**71.2%**
sarvam_gsm8k_indic (math)	Cross-EN	25.0%	32.5%	58.8%	36.2%	**68.8%**
indic_squad_odia (qa)	Odia	25.0%	26.2%	53.8%	27.5%	**53.8%**
indic_glue_wstp (multiple_choice)	Odia	25.0%	36.2%	36.2%	43.8%	**46.2%**
odia_truthfulqa_mc (multiple_choice)	Cross-EN	24.1%	32.5%	32.5%	31.2%	**36.2%**
Pure-Odia Macro Average (18 Tasks)	Odia	26.39%	41.60%	59.65%	43.40%	**60.83% (+34.44%)**
All-Tasks Macro Average (27 Tasks)	All	26.82%	38.74%	55.26%	40.02%	**56.83% (+30.01%)**

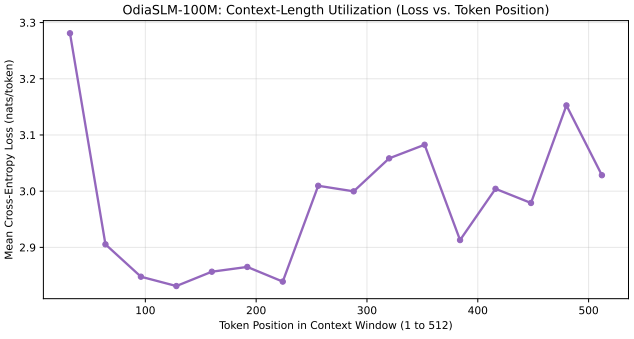


Figure 7: Context-length utilization curve for OdiaSLM-100M showing monotonic reduction in token prediction loss across positions 1 to 512.

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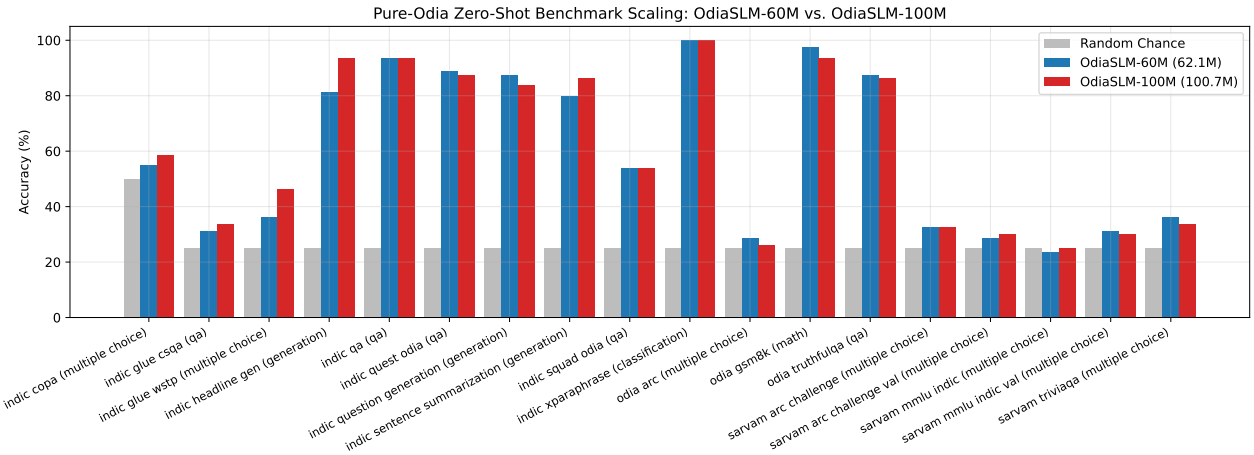


Figure 8: Zero-shot accuracy across the 18 Pure-Odia downstream benchmarks comparing Random Chance, OdiaSLM-60M, and OdiaSLM-100M.

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